

OpenVLA · arXiv 2406.09246 · CoRL 2024

Archaeologist

11+3 · seat 5

Discrete AR peer era + FAST upgrade

RT-2 → OpenVLA || Octo → π_0 → FAST → $\pi_{0.5}$

Oct 21 paper club · VLA Architectures

Spine: naïve discrete AR @ ~5-6 Hz · historical open AR baseline

BEHAVIOR deploy = $\pi_{0.5}$ / OpenPI / GR00T · OpenVLA = open AR control

Action-family taxonomy · CLUB SPINE

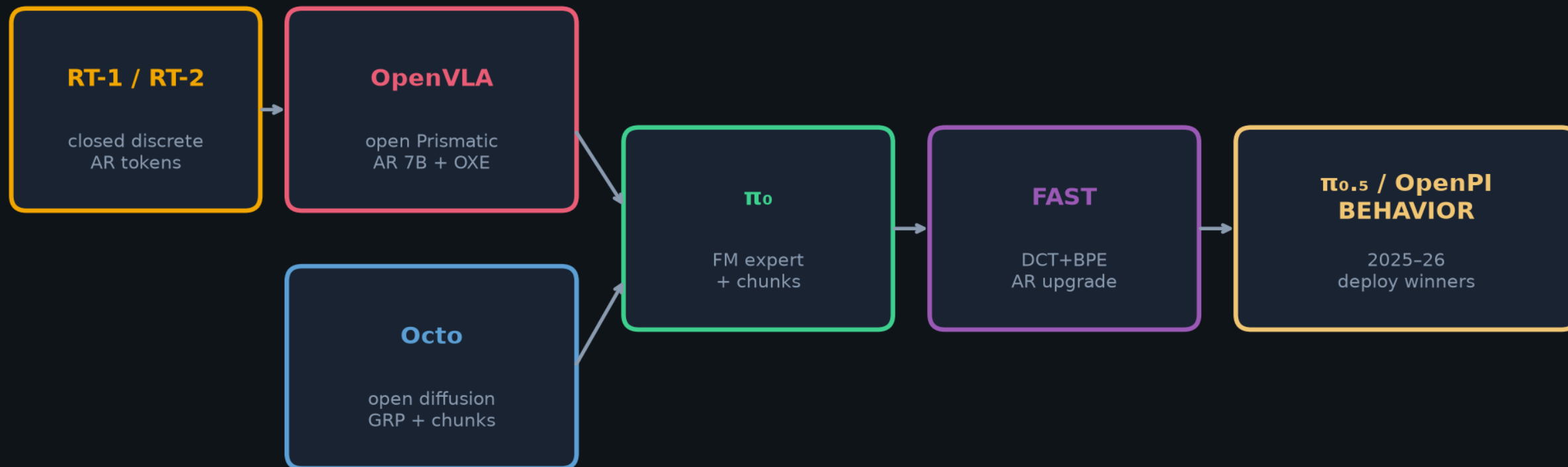
Must appear in every role · no-chunk AR $\approx$ 5-6 Hz

Family	Model	Signature
Naïve discrete AR bins	OpenVLA / RT-2	Lang grounding @ $\sim$ 5-6 Hz; weak high-Hz
DDPM chunks	Octo / DP	Smoother narrow skills
FM chunks	π_0 / $\pi_{0.5}$	2025-26 deploy winners
Compressed discrete AR	FAST / OpenVLA+FAST / π_0 -FAST	AR recovers high-Hz

Syllabus: OpenVLA = historical open AR baseline · BEHAVIOR deploy = $\pi_{0.5}$ / OpenPI / GROOT

Genealogy · Archaeologist MUST-HIT

RT-2-style discrete AR → OpenVLA ∥ Octo → π_0 → FAST → $\pi_{0.5}$ / OpenPI / BEHAVIOR



Peer era 2024: OpenVLA (AR VLM-VLA) ∥ Octo (diffusion GRP) ∥ closed RT-2-X

Same OXE water · different action families · FAST = AR improvement path, not FM retirement

Intellectual priors · Archaeologist

RT-1 / RT-2 / RT-2-X

Discrete AR VLA template; closed PaLI-scale foil

Prismatic VLMs

Open fused DINO+SigLIP backbone

Llama 2 · SigLIP · DINOv2

LM + dual vision towers

OXE / RT-X

Data pool + RT-1-X / RT-2-X baselines

Octo

Peer-era open OXE diffusion GRP; mix weights

Diffusion Policy

From-scratch IL + chunking foil in FT

BridgeData V2 · DROID

Eval / mix slices

LoRA · bitsandbytes

PEFT / serve story

Club one-liner

OpenVLA = open Llama/Prismatic discrete-AR VLA on OXE

Peer to Octo diffusion · ancestor contrast to π_0 FM

FAST is the AR patch · not 2026 BEHAVIOR SOTA

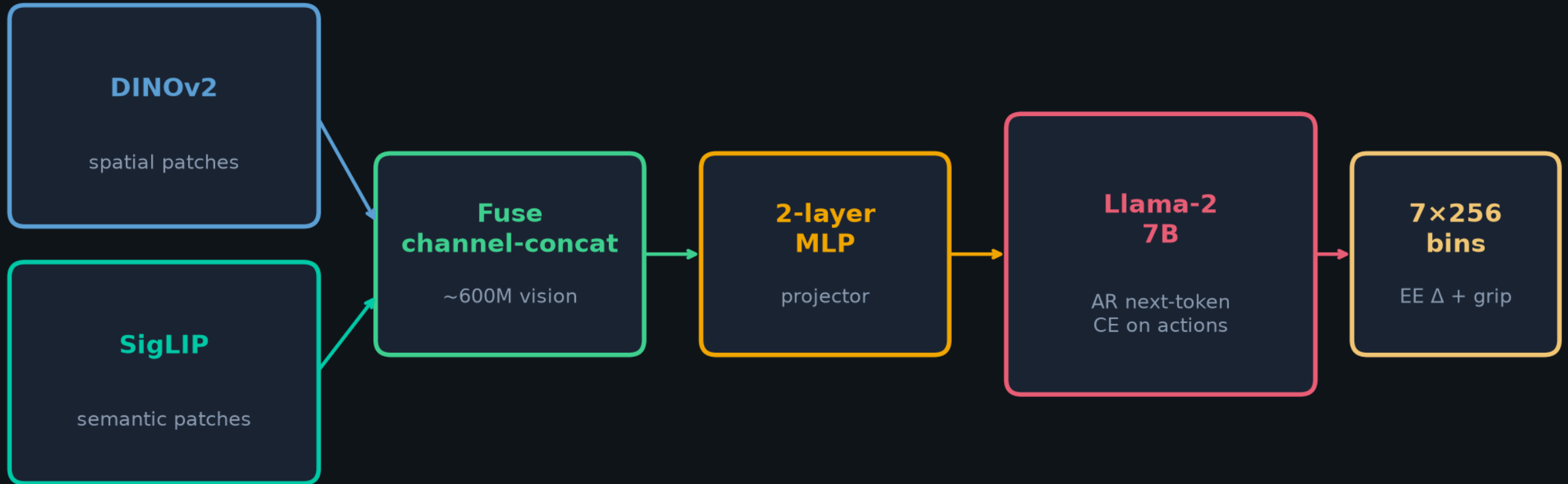
Stack: DINOv2||SigLIP + Llama-2 7B @ 224² · 256 bins · no chunk · ~6 Hz

Adoption unlock: LoRA $r=32$ on 1xA100 · int4 serve ~7 GB

GitHub title cue: [ECE 605] - OpenVLA Making Empiricist

OpenVLA architecture · Prismatic VLM → discrete AR actions

No wrist / proprio / history / chunks · single 3rd-person RGB 224²



Prompt: “What should the robot do to {task}? A:” + 7 action tokens

Train: OXE ~970k · 27 epochs · 64×A100 ×14d ≈ 21.5k A100-h · LR 2e-5 · vision UNFROZEN

AR discrete vs DP / FM continuous · Empiricist contrast

Axis	OpenVLA AR	DP / Octo	FM (π_0 family)
Action	7×256 bin ids	Denoise chunk	Flow on chunk
Temporal	Single step NO chunk	Chunk + TE/receding	Chunk + receding
Loss	Token CE	Score / ϵ -pred	FM / velocity
Infer cost	7 AR × 7B LM	K steps × small head	~10 NFE × ~300M
Hz	~6 (4090 bf16)	5-15+ w/ chunks	~30 challenge
Lang	STRONG (VLM NTP)	Needs lang encoder	VLM + task emb
Dexterity	Grid / choppy if slow	STRONG	STRONG
2025-26 role	Open ancestor / lang FT	Strong IL baseline	DEFAULT serve

Pick: open lang FT → OpenVLA LoRA · smooth high-Hz → DP/FM · BEHAVIOR'26 → $\pi_{0.5}$ + optional FAST aux

Fig.2 — architecture (DINOv2||SigLIP → Llama-2 → actions)

multi-task settings with multiple objects. Following these results, we are the first to demonstrate the effectiveness of compute-efficient fine-tuning methods leveraging low-rank adaptation [LoRA; 26] and model quantization [27] to facilitate adapting OpenVLA models on consumer-grade GPUs instead of large server nodes without compromising performance. As a final contribution, we open-source all models, deployment and fine-tuning notebooks, and the OpenVLA codebase for training VLAs at scale, with the hope that these resources enable future work exploring and adapting VLAs for robotics.

2 Related Work

Visually-Conditioned Language Models Visually-conditioned language models (VLMs), which are trained on Internet-scale data to generate natural language from input image(s) and language prompts, have been adopted for myriad applications from visual question answering [28–31] to object localization [32, 33]. One of the key advances fueling recent VLMs are model architectures that bridge features from pretrained vision encoders [8, 9, 25] with pretrained language models [10, 23, 34–36], directly building on advances in both computer vision and natural language modelling to create powerful multimodal models. While early work explored various architectures for cross-attending between vision and language features [37–41], new open-source VLMs [20, 42–44] have converged on a simpler “patch-as-token” approach, in which patch features from pretrained visual transformers are treated as tokens, and are then projected into the input space of a language model. This simplicity makes it easy to repurpose existing tools for training language models at scale for VLM training. We employ these tools in our work to scale VLA training, and specifically use VLMs from Karamcheti et al. [44] as our pretrained backbone, as they are trained from *multi-resolution visual features*, fusing low-level spatial information from DINOv2 [25] with higher-level semantics from SigLIP [9] to aid in visual generalization.

Generalist Robot Policies A recent trend in robotics works towards training multi-task “generalist” robot policies [2, 6, 45–49] on large diverse robot datasets [1, 2, 6, 11, 45, 49–56], spanning many different robot embodiments [1, 5, 53, 57–66]. Notably, Octo [5] trains a generalist policy that can control multiple robots out-of-the-box and allows for flexible fine-tuning to new robot setups. A key difference between these approaches and OpenVLA is the model architecture. Prior works like Octo typically compose pretrained components such as language embeddings or visual encoders with additional model components initialized from scratch [2, 5, 6], learning to “stitch” them together during the course of policy training. Unlike these works, OpenVLA adopts a more end-to-end approach, directly fine-tuning VLMs to generate robot actions by treating them as tokens in the language model vocabulary. Our experimental evaluation shows that this simple yet scalable pipeline substantially boosts performance and generalization ability over prior generalist policies.

Vision-Language-Action Models A number of works have explored the use of VLMs for robotics, e.g., for visual state representations [12, 13], object detection [67], high-level planning [16], and for providing a feedback signal [68–71]. Others integrate VLMs directly into end-to-end visuomotor manipulation policies [14, 15], but incorporate significant structure into the policy architecture or require calibrated cameras, which limits their applicability. A number of recent works have explored similar recipes to ours and directly fine-tuned large pretrained VLMs for predicting robot actions [1, 7, 17, 18, 72–74]. Such models are often referred to as vision-language-action models (VLAs), since they fuse robot control actions directly into VLM backbones. This has three key benefits: (1) it performs alignment of pretrained vision and language components on a large, Internet-scale vision-language dataset, (2) the use of a generic architecture, not custom-made for robot control, allows us to leverage the scalable infrastructure underlying modern VLM training [75–77] and scale to training billion-parameter policies with minimal code modifications, and (3) it provides a direct pathway for robotics to benefit from the rapid improvements in VLMs. Existing works on VLAs either focus on training and evaluating in single robot or simulated setups [72–74, 78] and thus lack generality, or are closed and do not support efficient fine-tuning to new robot setups [1, 7, 17, 18]. Most closely related, RT-2-X [1] trains a 55B-parameter VLA policy on the Open X-Embodiment dataset and demonstrates state-of-the-art generalist manipulation policy performance. However, our work differs from RT-2-X in multiple important aspects: (1) by combining a strong open VLM

Fig.1 — system card / OpenVLA overview

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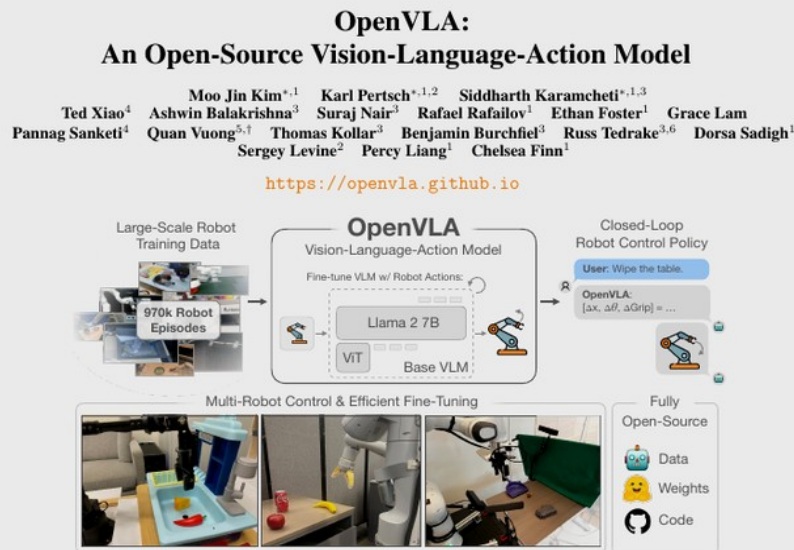


Figure 1: We present OpenVLA, a 7B-parameter open-source vision-language-action model (VLA), trained on 970k robot episodes from the Open X-Embodiment dataset [1]. OpenVLA sets a new state of the art for generalist robot manipulation policies. It supports controlling multiple robots out of the box and can be quickly adapted to new robot domains via parameter-efficient fine-tuning. The OpenVLA checkpoints and PyTorch training pipeline are fully open-source and models can be downloaded and fine-tuned from HuggingFace.

Abstract: Large policies pretrained on a combination of Internet-scale vision-language data and diverse robot demonstrations have the potential to change how we teach robots new skills: rather than training new behaviors from scratch, we can fine-tune such vision-language-action (VLA) models to obtain robust, generalizable policies for visuomotor control. Yet, widespread adoption of VLAs for robotics has been challenging as 1) existing VLAs are largely closed and inaccessible to the public, and 2) prior work fails to explore methods for efficiently fine-tuning VLAs for new tasks, a key component for adoption. Addressing these challenges, we introduce OpenVLA, a 7B-parameter open-source VLA trained on a diverse collection of 970k real-world robot demonstrations. OpenVLA builds on a Llama 2 language model combined with a visual encoder that fuses pretrained features from DINOv2 and SigLIP. As a product of the added data diversity and new model components, OpenVLA demonstrates strong results for generalist manipulation, outperforming closed models such as RT-2-X (55B) by 16.5% in absolute task success rate across 29 tasks and multiple robot embodiments, with 7x fewer parameters. We further show that we can effectively fine-tune OpenVLA for new settings, with especially

^{*}: denotes equal contribution

Correspondence to: moojink@stanford.edu, pertsch@berkeley.edu, skaramcheti@stanford.edu

¹Stanford University, ²UC Berkeley, ³Toyota Research Institute, ⁴Google Deepmind, ⁵Physical Intelligence,

⁶MIT, [†]Work done in part while at Google Deepmind

Fig.3 — WidowX Bridge bars (OpenVLA 70.6%)

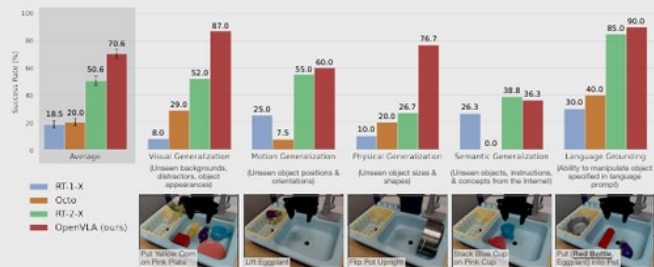


Figure 3: BridgeData V2 WidowX robot evaluation tasks and results. We evaluate OpenVLA and prior state-of-the-art generalist robot policies on a comprehensive suite of tasks covering several axes of generalization, as well as tasks that specifically assess language conditioning ability. OpenVLA achieves highest overall performance and even outperforms closed-source model RT-2-X in all categories except for semantic generalization. Average success rates $\pm$ StdErr are computed across 170 total rollouts per approach. See Table 4 for detailed results.

techniques for large transformer model training such as automatic mixed precision (AMP, PyTorch [75]), FlashAttention [76], and fully sharded data parallelism (FSDP, Zhao et al. [77]). Out of the box, the OpenVLA codebase has full support for training on the Open X dataset, integrates with HuggingFace’s [21] AutoModel class, and supports LoRA fine-tuning [26] and quantized model inference [27, 88].

5 Experiments

The goal of our experimental evaluations is to test OpenVLA’s ability to serve as a powerful multi-robot control policy out of the box, as well as be a good initialization for fine-tuning to new robot tasks. Concretely, we aim to answer the following questions:

1. How does OpenVLA compare to prior generalist robot policies, when evaluating on multiple robots and various types of generalization?
2. Can OpenVLA be effectively fine-tuned on a new robot setup and task, and how does it compare to state-of-the-art data-efficient imitation learning approaches?
3. Can we use parameter-efficient fine-tuning and quantization to reduce the computational requirements for training and inference of OpenVLA models and make them more accessible? What are the performance-compute trade-offs?

5.1 Direct Evaluations on Multiple Robot Platforms

Robot Setups and Tasks. We evaluate OpenVLA’s performance “out-of-the-box” on two robot embodiments: the WidowX robot from the BridgeData V2 evaluations [6] (see Fig. 1, left) and the mobile manipulation robot from the RT-1 and RT-2 evaluations [2, 7] (“Google robot”; see Fig. 1, middle). Both platforms have been extensively used in prior works for evaluating generalist robot policies [1, 2, 5, 7]. We define a comprehensive set of evaluation tasks in each environment that covers various axes of generalization, such as **visual** (unseen backgrounds, distractor objects, colors/appearances of objects); **motion** (unseen object positions/orientations); **physical** (unseen object sizes/shapes); and **semantic** (unseen target objects, instructions, and concepts from the Internet) generalization. We also assess language conditioning ability in scenes with multiple objects, testing whether the policy can manipulate the correct target object, as specified in the user’s prompt. See bottom row of Fig. 3 and Fig. 4 for example task images in the BridgeData V2 and Google robot evaluations, respectively. Overall, we evaluated each method in 170 rollouts (17 tasks with 10 trials each) for BridgeData V2 experiments and 60 rollouts (12 tasks with 5 trials each) for Google robot experiments. A detailed breakdown of all tasks and how they differ from the training data is in

Harness vs deploy · Visionary → 2026 BEHAVIOR

Skip Discord / Drive entirely

HARNESS · keep

- OpenVLA as bin-AR control
- LoRA / int4 recipes
- Language multi-object evals
- OXE-style mix morals
- Demo cleaning (no-ops!)

DEPLOY · ship

- $\pi_{0.5}$ / OpenPI / GR00T-class
- Chunked continuous FM
- Optional FAST aux (not primary)
- Multi-cam + proprio + stages
- Task/stage tokens for fixed-50

1 Do not ship pure OpenVLA AR as primary 30 Hz controller

2 If AR research continues → FAST/chunks before high-Hz claims

3 Steal LoRA recipe into FM stacks

4 Honesty: Bridge/Google % $\neq$ BEHAVIOR q-score

What aged well · durable OpenVLA gifts

Open VLM→VLA interface

Right community iteration surface

Fused DINO+SigLIP

Foreshadows multi-encoder stacks

PEFT/quant for 7B

Made VLAs lab-accessible

Authors' chunking call

Correctly predicted π_0 /DP winning recipe

DROP as default policy head: naïve per-step 256-bin AR without chunk compression

Archaeologist · close

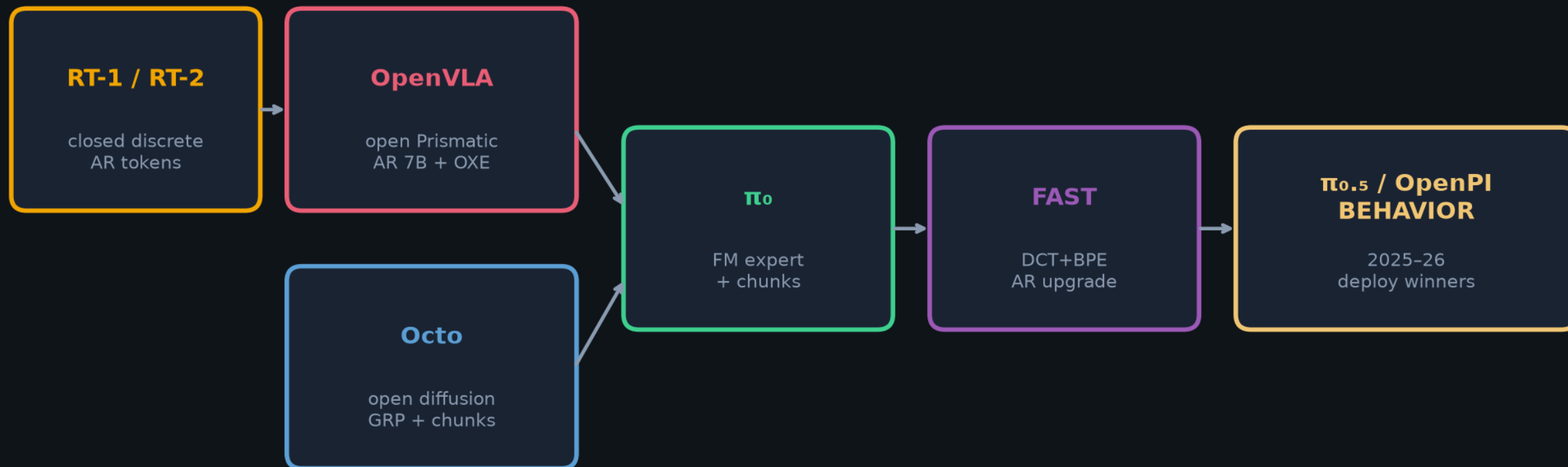
Genealogy: RT-2-style AR → OpenVLA ∥ Octo → π_0 → FAST → $\pi_{0.5}$ /OpenPI/BEHAVIOR.

Peer-era twin of Octo · ancestor contrast to FM deploy

Q&A · Oct 21 · OpenVLA / 2406.09246

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